MLE习题

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题目

- Basic Probability and MLE
- You are trapped in a dark cave with three indistinguishable exits on the walls. One of the exits takes you 3 hours to travel and takes you outside. One of the other exits takes 1 hour to travel and the other takes 2 hours, but both drop you back in the original cave. You have no way of marking which exits you have attempted. What is the expected time it takes for you to get outside?

解答

- Basic Probability and MLE
- Answer: Let the random variable, X be the time it takes for you to get outside. So, by the description of the problem,

$$E(X) = \frac{1}{3}(3) + \frac{1}{3}\,(1 + E(X)) + \frac{1}{3}(2 + E(X)).$$

- Solving this equation leads to the solution, $E(X) = 6$.

题目

- Basic Probability and MLE

- Let $X_1, \cdots, X_n$ be iid data from a uniform distribution over the disc of radius θ in R^2. Thus, $X_i \in \mathbb{R}^2$ and

$$p(x;\theta) = \begin{cases} \frac{1}{\pi\theta^2} & \text{if } \| x \| \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

- where $\| x \| = \sqrt{x_1^2 + x_2^2}$. Please find the maximum likelihood estimate of θ.

解答

- Basic Probability and MLE
- Answer: The likelihood function is

$$L(X^n;\theta) = \frac{1}{(\pi\theta^2)^n} 1\left\{\max_{1\leq i\leq n} \| X_i \| \leq \theta\right\}$$

- When θ increases, the likelihood decreases. So θ should be as small as possible. However, we also need that

$$\theta \geq \max'_{1\leq i\leq n} \| X_i \|$$

- otherwise the likelihood will drop to " 0." So the maximum likelihood estimator is

$$\hat{\theta} = \max_{1\leq i\leq n} \| X_i \|$$

EM算法习题

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判断题

- Suppose you are given an EM algorithm that finds maximum likelihood estimates for a model with latent variables. You are asked to modify the algorithm so that it finds MAP estimates instead. Which step or steps do you need to modify (B)
- A. Expectation B. Maximization C. No modification necessary D. Both

解答

- Solution:
- To modify the **EM algorithm** from finding **Maximum Likelihood Estimates (MLE)** to finding **Maximum A Posteriori (MAP)** estimates, you need to adjust the **Maximization (M)** step.
- **1. Maximum Likelihood Estimation (MLE):**
- In the standard EM algorithm for MLE, the goal is to maximize the likelihood function$P(X|\theta)$, where$x$represents the observed data and θ are the model parameters.
- The likelihood function is maximized as:

$$\theta^{\text{MLE}} = \operatorname{argmax}_{\theta} \log P(X|\theta).$$

解答

- Solution:
- **2. Maximum A Posteriori Estimation (MAP):**
- In MAP estimation, you introduce a **prior distribution** $P(\theta)$ over the parameters θ. The objective is to maximize the **posterior distribution** $P(\theta|X)$, which is given by Bayes' rule:

$$P(\theta|X) \propto P(X|\theta)P(\theta).$$

- Taking the **logarithm**, the optimization problem becomes:

$$\theta^{\text{MAP}} = \text{argmax}_{\theta}[\log P(X|\theta) + \log P(\theta)].$$

Here, $\log P(X|\theta)$ is the data likelihood, and $\log P(\theta)$ represents the prior.

解答

- Solution:
- **3. Modifications to the EM Algorithm:**
- **• Expectation (E) Step:**
- The E–step computes the expectation of the latent variables conditioned on the current parameter estimates θ. This step does not depend on the prior $P(\theta)$, so it remains **unchanged**.
- **• Maximization (M) Step:**
- The M–step, which previously maximized only the likelihood $\log P(X|\theta)$, must now include the prior $\log P(\theta)$.
- The updated M–step maximizes the posterior instead:

$$\theta^{\text{new}} = \text{argmax}_{\theta}[\mathbb{E}_{Z|X,\theta}\text{old}[\log P(X,Z|\theta)] + \log P(\theta)],$$

- where Z are the latent variables.

判断题

- The EM algorithm optimizes a lower bound on its objective function, which is the marginal likelihood $\prod_i P(x_i)$ of the observed data points $x_1, x_2, \ldots x_N$

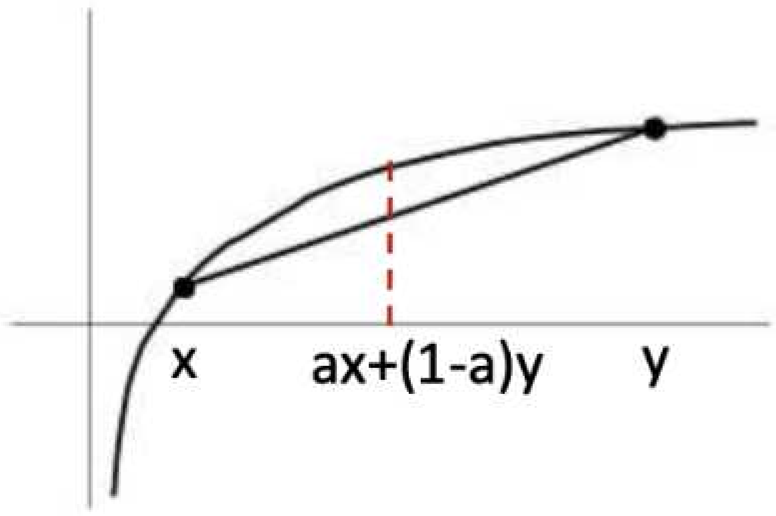
解答

- True
- Lower-bound on marginal likelihood

$$P(D;\theta) = \sum_{j=1}^{m} \log \sum_{\mathbf{z}} P\left(\mathbf{x}_j, \mathbf{z}|\theta\right)$$

$$= \sum_{j=1}^{m} \log \sum_{\mathbf{z}} \underbrace{Q\left(\mathbf{z}|\mathbf{x}_j\right)}_{P(z)} \underbrace{\frac{P\left(\mathbf{z}, \mathbf{x}_j|\theta\right)}{Q\left(\mathbf{z}|\mathbf{x}_j\right)}}_{f(z)}$$

Jensen's inequality $\log \sum_{\mathbf{z}} P(\mathbf{z})\, f(\mathbf{z}) \geq \sum_{\mathbf{z}} P(\mathbf{z}) \log f(\mathbf{z})$



log: concave function

$\log(ax+(1-a)y) \geq a \log(x) + (1-a) \log(y)$

判断题

The EM algorithm is guaranteed to never decrease the value of its objective function on any iteration.

解答

- True
- 由EM算法的收敛性可得，我们在课堂上证明过

判断题

The objective function optimized by the EM algorithm can also be optimized by a gradient descent algorithm which will find the global optimal solution, whereas EM finds its solution more quickly but may return only a locally optimal solution.

解答

- False. The only false part is that the gradient descent algorithm will find the global optimal solution. Gradient descent can also get stuck in a local optima.

计算题

- Consider data, $D = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ * \end{pmatrix} \right\}$ two-dimensional(separable) distribution $p(x_1, x_2) = p_{x_1}(x_1) p_{x_2}(x_2)$, with

$$p_{x_1}(x_1) = \begin{cases} \frac{1}{\theta_1} e^{-x_1/\theta_1} & \text{if } x_1 \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad p_{x_2}(x_2) = \begin{cases} \frac{1}{\theta_2} & \text{if } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases}$$

and a missing feature value, *.

a. What can you infer from θ2 by looking at D?

解答

- Consider data, $D = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ * \end{pmatrix} \right\}$ two-dimensional(separable) distribution $p(x_1, x_2) = p_{x_1}(x_1) p_{x_2}(x_2)$, with

$$p_{x_1}(x_1) = \begin{cases} \frac{1}{\theta_1} e^{-x_1/\theta_1} & \text{if } x_1 \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad p_{x_2}(x_2) = \begin{cases} \frac{1}{\theta_2} & \text{if } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases}$$

and a missing feature value, *. $x_{22} = 5 \leq \theta_2$.

a. What can you infer from θ2 by looking at D?

Solution: θ2 is necessarily greater or equal to 5 because

$$0 < x_{12} = 3 < x_{22} = 5 \leq \theta_2.$$

计算题

- Consider data, $D = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ * \end{pmatrix} \right\}$ two-dimensional(separable) distribution $p(x_1, x_2) = p_{x_1}(x_1)p_{x_2}(x_2)$, with

$$p_{x_1}(x_1) = \begin{cases} \frac{1}{\theta_1} e^{-x_1/\theta_1} & \text{if } x_1 \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad p_{x_2}(x_2) = \begin{cases} \frac{1}{\theta_2} & \text{if } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases}$$

and a missing feature value, *.

b. Start with an initial estimate $\underline{\theta}^0 = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ and analytically calculate $Q(\underline{\theta}; \underline{\theta}^0)$

- This is the estimate step in the EM algorithm.

解答

- Consider data, $D = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ * \end{pmatrix} \right\}$ two-dimensional(separable) distribution $p(x_1, x_2) = p_{x_1}(x_1) p_{x_2}(x_2)$, with

$$p_{x_1}(x_1) = \begin{cases} \frac{1}{\theta_1} e^{-x_1/\theta_1} & \text{if } x_1 \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad p_{x_2}(x_2) = \begin{cases} \frac{1}{\theta_2} & \text{if } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases}$$

and a missing feature value, *.

- Solution： The probability of one sample can be determined by:

$$\begin{aligned} p(\mathbf{x}|\underline{\theta}) = p(x_1, x_2|\underline{\theta}) &= p_{x1}(x_1|\underline{\theta}) p_{x_2}(x_2|\underline{\theta}) \\ &= \begin{cases} \left(\frac{1}{\theta_1} e^{-x_1/\theta_1}\right)\left(\frac{1}{\theta_2}\right) & \text{if } x_1 \geq 0 \text{ and } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{\theta_1 \theta_2} e^{-x_1/\theta_1} & \text{if } x_1 \geq 0 \text{ and } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

解答

Formulating the log likelihood of the data and marginalizing over the possible values of the unknown data:

$$
\begin{aligned}
Q(\underline{\theta};\underline{\theta}^0) &= E_{x_{32}}[\ln p(\mathbf{x}_g,\mathbf{x}_b;\underline{\theta})|\underline{\theta}^0;D_g] \\
&= \int_{-\infty}^{\infty}\left[\sum_{k=1}^{2}\ln p(\mathbf{x}_k|\underline{\theta}) + \ln p(\mathbf{x}_3|\underline{\theta})\right] p(x_{32}|x_{31}=2;\underline{\theta}^0)dx_{32} \\
&= \sum_{k=1}^{2}\ln p(\mathbf{x}_k|\underline{\theta}) + \int_{-\infty}^{\infty}\ln p\left(\left[\begin{array}{c}2\\ x_{32}\end{array}\right]|\underline{\theta}\right)\frac{p\left(\left[\begin{array}{c}2\\ x_{32}\end{array}\right]|\underline{\theta}^0\right)}{\int_{-\infty}^{\infty}p\left(\left[\begin{array}{c}2\\ x'_{32}\end{array}\right]|\underline{\theta}^0\right)dx'_{32}}dx_{32}
\end{aligned}
$$

解答

Note that x_31 and x_32 are independent with each other. Thus,

$$p\left(\left[\begin{array}{c}2\\x_{32}\end{array}\right]|\underline{\theta}^0\right)=p_{x_1}\left(x_{31}=2|\underline{\theta}^0\right)p_{x_2}(x_{32}|\underline{\theta}^0),$$

$$\int_{-\infty}^{\infty}p\left(\left[\begin{array}{c}2\\x'_{32}\end{array}\right]|\underline{\theta}^0\right)dx'_{32}=p_{x_1}\left(x_{31}=2|\underline{\theta}^0\right)\int_{-\infty}^{\infty}p_{x_2}(x'_{32}|\underline{\theta}^0)dx'_{32}=p_{x_1}\left(x_{31}=2|\underline{\theta}^0\right)$$

Therefore,

$$Q(\underline{\theta};\underline{\theta}^0)=\sum_{k=1}^{2}\ln p(\mathbf{x}_k|\underline{\theta})+\int_{-\infty}^{\infty}\ln p\left(\left[\begin{array}{c}2\\x_{32}\end{array}\right]|\underline{\theta}\right)p_{x_2}(x_{32}|\underline{\theta}^0)dx_{32}$$

解答

Note that $p\left(\begin{bmatrix} 2 \\ x_{32} \end{bmatrix} | \underline{\theta}\right)$ is different than zero if $0 \le x_{32} \le \theta_2$;

and $p_{x_2}(x_{32}|\underline{\theta}^0)$ is different than zero if $0 \le x_{32} \le \theta_2^0 = 6$.

Therefore, $p\left(\begin{bmatrix} 2 \\ x_{32} \end{bmatrix} | \underline{\theta}\right) p_{x_2}(x_{32}|\underline{\theta}^0)$ is different than zero between [0, $\min(\theta_2 \ , 6)$]

$$\sum_{k=1}^{2} \ln p(\mathbf{x}_k|\underline{\theta}) = \ln \frac{1}{\theta_1\theta_2} e^{-1/\theta_1} + \ln \frac{1}{\theta_1\theta_2} e^{-4/\theta_1} = 2\ln \frac{1}{\theta_1\theta_2} - \frac{5}{\theta_1}$$

$$\int_{-\infty}^{\infty} \ln p\left(\begin{bmatrix} 2 \\ x_{32} \end{bmatrix} | \underline{\theta}\right) p_{x_{32}}(x_{32}|\underline{\theta}^0) dx_{32} = \int_0^{\min(\theta_2,6)} \left(\ln \frac{1}{\theta_1\theta_2} e^{-2/\theta_1}\right)\left(\frac{1}{6}\right) dx_{32}$$

$$= \begin{cases} \theta_2 \frac{1}{6}\left(\ln \frac{1}{\theta_1\theta_2} - \frac{2}{\theta_1}\right) & \text{if } \min(\theta_2, 6) = \theta_2, \quad \text{where } 5 \le \theta_2 < 6 \\ \ln \frac{1}{\theta_1\theta_2} - \frac{2}{\theta_1} & \text{if } \min(\theta_2, 6) = 6, \quad \text{where } \theta_2 \ge 6 \end{cases}$$

解答

$$Q(\underline{\theta};\underline{\theta}^0) = \sum_{k=1}^{2} \ln p(\mathbf{x}_k|\underline{\theta}) + \int_{-\infty}^{\infty} \ln p\left(\left[\begin{array}{c} 2 \\ x_{32} \end{array}\right]|\underline{\theta}\right) p_{x_{32}}\left(x_{32}|\underline{\theta}^0\right) dx_{32}$$

$$= \begin{cases} 2\ln\frac{1}{\theta_1\theta_2} - \frac{5}{\theta_1} + \frac{1}{6}\theta_2\left(\ln\frac{1}{\theta_1\theta_2} - \frac{2}{\theta_1}\right) & \text{where } 5 \le \theta_2 < 6 \\ 3\ln\frac{1}{\theta_1\theta_2} - \frac{7}{\theta_1} & \text{where } \theta_2 \ge 6 \end{cases}$$

计算题

- Consider data, $D=\left\{\begin{pmatrix}1\\3\end{pmatrix},\begin{pmatrix}4\\5\end{pmatrix},\begin{pmatrix}2\\ *\end{pmatrix}\right\}$ two-dimensional(separable) distribution $p(x_1,x_2)=p_{x_1}(x_1)p_{x_2}(x_2)$, with

$$p_{x_1}(x_1)=\begin{cases}\frac{1}{\theta_1}e^{-x_1/\theta_1} & \text{if } x_1\geq 0\\ 0 & \text{otherwise}\end{cases}\qquad p_{x_2}(x_2)=\begin{cases}\frac{1}{\theta_2} & \text{if } 0\leq x_2\leq\theta_2\\ 0 & \text{otherwise}\end{cases}$$

and a missing feature value, *.

c. Find the $\underline{\theta}$ that maximizes your $Q(\underline{\theta};\underline{\theta}^0)$– the maximization step of the EM algorithm.

解答

- Consider data, $D = \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ * \end{pmatrix} \right\}$ two-dimensional(separable) distribution $p(x_1, x_2) = p_{x_1}(x_1) p_{x_2}(x_2)$, with

$$p_{x_1}(x_1) = \begin{cases} \frac{1}{\theta_1} e^{-x_1/\theta_1} & \text{if } x_1 \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad p_{x_2}(x_2) = \begin{cases} \frac{1}{\theta_2} & \text{if } 0 \leq x_2 \leq \theta_2 \\ 0 & \text{otherwise} \end{cases}$$

and a missing feature value, *.

Solution:

$$\frac{\partial Q(\underline{\theta}; \underline{\theta}^0)}{\partial \theta_2} = \begin{cases} -\frac{2}{\theta_2} - \frac{1}{6}\left(\frac{2}{\theta_1} - \ln \frac{1}{\theta_1 \theta_2}\right) - \frac{1}{6} & \text{where } 5 \leq \theta_2 < 6 \\ -\frac{3}{\theta_2} & \text{where } \theta_2 \geq 6 \end{cases}$$

解答

- Thus,$Q(\underline{\theta};\underline{\theta}^0) < 0$ where $\theta_2 \geq 5$. (Note $\frac{2}{\theta_1} - \ln\frac{1}{\theta_1\theta_2} = \frac{1}{\theta_1} + \left(\frac{1}{\theta_1} - \ln\frac{1}{\theta_1}\right) + \ln\theta_2 > 0$ because $\frac{1}{\theta_1} > 0$, $\left(\frac{1}{\theta_1} - \ln\frac{1}{\theta_1}\right) \geq 1$ and $\ln\theta_2 > 0$ where $\theta_1 > 0$ and $\theta_2 \geq 5$.)
- Second, $Q(\underline{\theta};\underline{\theta}^0)$ is continuous at $\theta_2 = 6$, because the two equations corresponding to the two intervals $(5 \leq \theta_2 < 6, \theta_2 \geq 6)$ have the same value at $\theta_2 = 6$, which is
- $3\ln\frac{1}{6\theta_1} - \frac{7}{\theta_1}$.

解答

- From these two facts, we can conclude that $Q(\underline{\theta};\underline{\theta}^0)$ is a monotonically decreasing continuous function with respect to θ_2. Thus, the maximum occurs when $\theta_2 = 5$.

- To find the value of θ_1 that maximizes Q when $\theta_2 = 5$,

- $\frac{\partial Q(\underline{\theta};\underline{\theta}^0)}{\partial \theta_1} = -\frac{2}{\theta_1} + \frac{5}{\theta_1^2} - \frac{5}{6\theta_1} + \frac{10}{6\theta_1^2} = 0. \qquad (1)$

- Thus, we find that

- $\mathrm{argmax}_{\theta_1} Q(\underline{\theta};\underline{\theta}^0) = \frac{40}{17}$

- (i.e., $\theta_1 = \frac{40}{17}$ and $\theta_2 = 5$).

计算题

- **An EM algorithm for a Mixture of Bernoullis**
- In this section, you will derive an expectation-maximization (EM) algorithm to cluster black and white images. The inputs $x^{(i)}$ can be thought of as vectors of binary values corresponding to black and white pixel values, and the goal is to cluster the images into groups. You will be using a mixture of Bernoullis model to tackle this problem.
- For the sake of brevity, you do not need to substitute in previously derived expression in later problems.
- For example, beyond question 1.1.1 you may use $P(x^{(i)}|p^{(k)})$ in your answers.

计算题

- **An EM algorithm for a Mixture of Bernoullis**

1.1 Mixture of Bernoullis

Consider a vector of binary random variables, $x \in \{0,1\}^D$. Assume each variable x_d is drawn from a Bernoulli(p_d) distribution, so $P(x_d = 1) = p_d$. Let $p \in (0,1)^D$ be the resulting vector of Bernoulli parameters. Write an expression for $P(x|p)$.

解答

- **An EM algorithm for a Mixture of Bernoullis**

1.1 Mixture of Bernoullis

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- **Solution:**

- $P(x|p) = \prod_{d=1}^{D} p_d^{x_d} (1 - p_d)^{1-x_d} \qquad (1)$

计算题

- **An EM algorithm for a Mixture of Bernoullis**

  1.1 Mixture of Bernoullis

- Now suppose we have a mixture of K Bernoulli distributions: each vector $x^{(i)}$ is drawn from some vector of Bernoulli random variables with parameters $p^{(k)}$, we will call this Bernoulli$(p^{(k)})$. Let $\{p^{(1)}, \ldots, p^{(K)}\} = p$. Assume a distribution $\pi^{(k)}$ over the selection of which set of Bernoulli parameters $p^{(k)}$ is chosen. Write an expression for $P(x^{(i)}|p, \pi)$.

解答

- **An EMalgorithm for a Mixture of Bernoullis**

  1.1 Mixture of Bernoullis

- Now suppose we have a mixture of K Bernoulli distributions: each vector $x^{(i)}$ is drawn from some vector of Bernoulli random variables with parameters $p^{(k)}$, we will call this Bernoulli$(p^{(k)})$. Let $\{p^{(1)}, \ldots, p^{(K)}\} = p$. Assume a distribution $\pi^{(k)}$ over the selection of which set of Bernoulli parameters $p^{(k)}$ is chosen. Write an expression for $P(x^{(i)}|p, \pi)$.

- **Solution:**

  Let A_k be the event that $x = x^{(i)}$ was drawn from $p^{(k)}$. Then:

- $P(x|p,\pi) = \sum_k P(x, A_k|p,\pi) = \sum_k P(x|A_k, p, \pi)P(A_k|p,\pi) = \sum_k \pi_k P(x|p^{(k)})$ (2)

计算题

- **An EM algorithm for a Mixture of Bernoullis**

  1.1 Mixture of Bernoullis

- Finally, suppose we have inputs $X = \{x^{(i)}\}_{i=1}^{n}$. Using the above, write an expression for the log-likelihood of the data X, $\log P(X|\pi, p)$.

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.1 Mixture of Bernoullis

- Finally, suppose we have inputs $X = \{x^{(i)}\}_{i=1}^{n}$. Using the above, write an expression for the log-likelihood of the data X, $\log P(X|\pi, p)$.

- **Solution**:

- $\log L(p, \pi) = \sum_{i=1}^{N} \log P(x^{(i)}|p, \pi)$ (3)

计算题

- **An EM algorithm for a Mixture of Bernoullis**

  1.2 Expectation Step

- Now, we introduce the latent variables for the EM algorithm. Let $z^{(i)} \in \{0,1\}^K$ be an indicator vector, such that $z_k^{(i)} = 1$ if $x^{(i)}$ was drawn from a Bernoulli$(p^{(k)})$, and 0 otherwise. Let $Z = \{z^{(i)}\}_{i=1}^n$. What is $P(z^{(i)}|\pi)$? What is $P(x^{(i)}|z^{(i)}, p, \pi)$?

解答

- **An EM algorithm for a Mixture of Bernoullis**

1.2 Expectation Step

Now, we introduce the latent variables for the EM algorithm. Let $z^{(i)} \in \{0,1\}^K$ be an indicator vector, such that $z_k^{(i)} = 1$ if $x^{(i)}$ was drawn from a Bernoulli$(p^{(k)})$, and 0 otherwise. Let $Z = \{z^{(i)}\}_{i=1}^n$. What is $P(z^{(i)}|\pi)$? What is $P(x^{(i)}|z^{(i)}, p, \pi)$?

- **Solution**:

Let $A_k^{(i)}$ be the event that $x^{(i)}$ was drawn from $p^{(k)}$.

- $P(z^{(i)}|\pi) = \prod_{k=1}^{K} \pi_k^{z_k^{(i)}}$ (4)

- $P(x^{(i)}|z^{(i)}, p, \pi) = \prod_{k=1}^{K} [P(x^{(i)}|p^{(k)})]^{z_k^{(i)}}$ (5)

计算题

- **An EM algorithm for a Mixture of Bernoullis**

1.2 Expectation Step

Using the above two quantities, derive the likelihood of the data and the latent variables, $P(Z, X|\pi, p)$.

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.2 Expectation Step

  Using the above two quantities, derive the likelihood of the data and the latent variables, $P(Z, X|\pi, p)$.

- **Solution:**

- $P(Z, X|\pi, p) = \prod_{i=1}^{N} P\left(x^{(i)}, z^{(i)}|\pi, p\right) = \prod_{i=1}^{N} P\left(x^{(i)}|z^{(i)}, \pi, p\right)P\left(z^{(i)}|\pi\right) \qquad (6)$

- $= \prod_{i=1}^{N}\left[\prod_{k=1}^{K}\left[P\left(x^{(i)}|p^{(k)}\right)\right]^{z_k^{(i)}}\right]\left[\prod_{k=1}^{K}\pi_k^{z_k^{(i)}}\right] \qquad (7)$

计算题

- **An EM algorithm for a Mixture of Bernoullis**

  1.2 Expectation Step

- Let $\eta\left(z_k^{(i)}\right) = E\left[z_k^{(i)} | x^{(i)}, \pi, p\right]$. Show that:

- $$\eta\left(z_k^{(i)}\right) = \frac{\pi_k \prod_{d=1}^{D} \left(p_d^{(k)}\right)^{x_d^{(i)}} \left(1 - p_d^{(k)}\right)^{1-x_d^{(i)}}}{\sum_j \pi_j \prod_{d=1}^{D} \left(p_d^{(j)}\right)^{x_d^{(i)}} \left(1 - p_d^{(j)}\right)^{1-x_d^{(i)}}}. \qquad (8)$$

- Let $\tilde{p}$, $\tilde{\pi}$ be the new parameters that we'd like to maximize, so p, π are from the previous iteration. Use this to derive the following final expression for the E step in the expectation-maximization algorithm:

- $$E[\log P(Z, X | \tilde{p}, \tilde{\pi}) | X, p, \pi] = \sum_{i=1}^{N} \sum_{k=1}^{K} \eta\left(z_k^{(i)}\right) \left[\log \tilde{\pi}_k + \sum_{d=1}^{D} \left(x_d^{(i)} \log \tilde{p}_d^{(k)} + (1 - x_d^{(i)}) \log\left(1 - \tilde{p}_d^{(k)}\right)\right)\right]$$

解答

An EM algorithm for a Mixture of Bernoullis

1.2 Expectation Step

- We begin by expressing $\eta\left(z_k^{(i)}\right)$:$\eta\left(z_k^{(i)}\right) = E\left[z_k^{(i)} | x^{(i)}, \pi, p\right] = P\left[z_k^{(i)} = 1 | x^{(i)}, \pi, p\right].$ (10)
- Using Bayes' rule, we get: $\eta\left(z_k^{(i)}\right) = \frac{P\left(x^{(i)} | z_k^{(i)} = 1, p^{(k)}\right) P\left(z_k^{(i)} = 1 | \pi\right)}{\sum_j P\left(x^{(i)} | z_j^{(i)} = 1, p^{(j)}\right) P\left(z_j^{(i)} = 1 | \pi\right)}.$ (11)
- Substituting the probabilities:
- $P\left(z_k^{(i)} = 1 | \pi\right) = \pi_k,$
- $P\left(x^{(i)} | z_k^{(i)} = 1, p^{(k)}\right) = \prod_{d=1}^{D} \left(p_d^{(k)}\right)^{x_d^{(i)}} \left(1 - p_d^{(k)}\right)^{1 - x_d^{(i)}},$
- we get:
- $\eta\left(z_k^{(i)}\right) = \frac{\pi_k \prod_{d=1}^{D} \left(p_d^{(k)}\right)^{x_d^{(i)}} \left(1 - p_d^{(k)}\right)^{1 - x_d^{(i)}}}{\sum_j \pi_j \prod_{d=1}^{D} \left(p_d^{(j)}\right)^{x_d^{(i)}} \left(1 - p_d^{(j)}\right)^{1 - x_d^{(i)}}}.$ (12)

解答

An EM algorithm for a Mixture of Bernoullis

1.2 Expectation Step

- Next, we compute the log-likelihood of the complete data:
- $\log P(Z, X|\pi, p) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_k^{(i)} \left[\log \pi_k + \log \prod_{d=1}^{D} \left(p_d^{(k)}\right)^{x_d^{(i)}} \left(1 - p_d^{(k)}\right)^{1-x_d^{(i)}} \right].$ (13)
- Simplifying the logarithm of the product, we have:
- $\log P(Z, X|\pi, p) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_k^{(i)} \left[\log \pi_k + \sum_{d=1}^{D} \left(x_d^{(i)} \log p_d^{(k)} + \left(1 - x_d^{(i)}\right) \log\left(1 - p_d^{(k)}\right) \right) \right].$ (14)
- Taking the expected value with respect to $z_k^{(i)}$ and replacing $E\left[z_k^{(i)}\right] = \eta\left(z_k^{(i)}\right)$, finishes the solution.

计算题

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- We need to maximize the above expression with respect to $\tilde{\pi}, \tilde{p}$.

  First, show that the value of $\tilde{p}$ that maximizes the E step is:

- $$\tilde{p}_d^{(k)} = \frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right) x_d^{(i)}}{N_k}, \qquad (15)$$

- where $N_k = \sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)$.

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution**:

  To maximize with respect to $\tilde{p}_d^{(k)}$, we take the derivative:

- $$\frac{\partial}{\partial \tilde{p}_d^{(k)}} E[\log P(Z, X|\tilde{p}, \tilde{\pi})] = \sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)\left[\frac{x_d^{(i)}}{\tilde{p}_d^{(k)}} - \frac{1-x_d^{(i)}}{1-\tilde{p}_d^{(k)}}\right]. \qquad (16)$$

- Setting this derivative to 0 and simplifying leads to:

- $$\tilde{p}_d^{(k)} = \frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right) x_d^{(i)}}{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)} = \frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right) x_d^{(i)}}{N_k}. \qquad (17)$$

计算题

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- Show that the value of $\tilde{\pi}$ that maximizes the E step is:

- $$\tilde{\pi}_k = \frac{N_k}{\sum_{k'} N_{k'}}. \qquad (18)$$

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution**:

  To show that the value of $\tilde{\pi}$ that maximizes the E step is:

- $\tilde{\pi}_k = \frac{N_k}{\sum_{k'} N_{k'}}, \qquad (19)$

- where $N_k = \sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)$, we proceed as follows:

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution**:

  Step 1: Objective Function

- From the E step, the expression we need to maximize with respect to $\tilde{\pi}$ is:

- $E[\log P(Z,X|\tilde{\pi})] = \sum_{i=1}^{N}\sum_{k=1}^{K}\eta\left(z_k^{(i)}\right)\log\tilde{\pi}_k.$ (20)

- Here:

• $\eta\left(z_k^{(i)}\right)$ is the responsibility that component k has for data point $x^{(i)}$, which acts as a weight.

• $\tilde{\pi}_k$ are the mixture weights we wish to optimize.

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution:**

  Step 2: Constraint

- The mixture weights $\tilde{\pi}_k$ must satisfy the normalization constraint:

- $\sum_{k=1}^{K} \tilde{\pi}_k = 1.$ (21)

- To enforce this, we introduce a Lagrange multiplier λ and define the Lagrangian:

- $\mathfrak{L}(\tilde{\pi}, \lambda) = \sum_{i=1}^{N} \sum_{k=1}^{K} \eta\left(z_k^{(i)}\right) \log \tilde{\pi}_k + \lambda\left(1 - \sum_{k=1}^{K} \tilde{\pi}_k\right).$ (22)

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution:**

  Step 3: Take the Derivative

- To find the optimal $\tilde{\pi}_k$, we take the derivative of $\mathfrak{l}(\tilde{\pi}, \lambda)$ with respect to $\tilde{\pi}_k$:

- $$\frac{\partial \mathfrak{l}}{\partial \tilde{\pi}_k} = \frac{\partial}{\partial \tilde{\pi}_k}\left(\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right) \log \tilde{\pi}_k + \lambda\left(1 - \sum_{k=1}^{K} \tilde{\pi}_k\right)\right). \qquad (23)$$

- Simplify the derivative:

- $$\frac{\partial \mathfrak{l}}{\partial \tilde{\pi}_k} = \frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)}{\tilde{\pi}_k} - \lambda. \qquad (24)$$

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution**:

  Step 4: Set the Derivative to Zero

- To maximize the Lagrangian, set the derivative to zero:

- $$\frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)}{\tilde{\pi}_k} - \lambda = 0. \qquad (25)$$

- Rearranging for $\tilde{\pi}_k$, we get:

- $$\tilde{\pi}_k = \frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)}{\lambda}. \qquad (26)$$

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution:**

  Step 5: Solve for λ

- To determine λ, we use the normalization constraint $\sum_{k=1}^{K} \tilde{\pi}_k = 1$:
- $\sum_{k=1}^{K} \tilde{\pi}_k = \sum_{k=1}^{K} \frac{\sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)}{\lambda} = 1. \qquad (27)$
- Simplify: $\frac{1}{\lambda} \sum_{k=1}^{K} \sum_{i=1}^{N} \eta\left(z_k^{(i)}\right) = 1. \qquad (28)$
- Define $N_k = \sum_{i=1}^{N} \eta\left(z_k^{(i)}\right)$. Then:
- $\frac{1}{\lambda} \sum_{k=1}^{K} N_k = 1. \qquad (29)$
- Solving for λ, we find: $\lambda = \sum_{k=1}^{K} N_k . \qquad (30)$

解答

- **An EM algorithm for a Mixture of Bernoullis**

  1.3 Maximization Step

- **Solution:**

  Step 6: Final Solution

- Substitute λ back into the expression for $\tilde{\pi}_k$:

- $$\tilde{\pi}_k = \frac{N_k}{\lambda} = \frac{N_k}{\sum_{k'=1}^{K} N_{k'}}. \qquad (31)$$

- **Conclusion**

- The value of $\tilde{\pi}_k$ that maximizes the E step is:

- $$\tilde{\pi}_k = \frac{N_k}{\sum_{k'} N_{k'}}. \qquad (32)$$

参考文献

- CMU 10-701
- Stanford CS229
- 统计学习方法
- CMU 10-715
- UTORONTO CSC2515

End